

WHERE WEB AGENTS GO BLIND: DECOMPOSING REACHABILITY FROM TASK SUCCESS IN STRUCTURALLY CONTAINED WEB CONTENT

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ABSTRACT

Web-agent task failure can reflect missing evidence or errors after observation. We introduce a controlled benchmark separating *reachability* from task success across iframe depth and origin, shadow-Document Object Model (DOM) containment, and DOM-versus-canvas rendering. A 1,760-trial primary grid crosses four observation architectures with two base models on two fixed tasks. All architectures reached every non-canvas target. On canvas, DOM, accessibility-tree, and Chrome DevTools Protocol observations fell from 40/40 to 0/40 reachability; screenshots remained at 40/40, but the original vision deployments completed 0/40 trials because they appended an adjacent marker. Two added deployments revealed model heterogeneity: `gpt-5` completed 15/20 original-layout canvas trials and `gpt-5-mini` 0/20. A 2,640-trial revision extends separation sweeps, measures pixel and DOM separation in familiar layouts, tests boundary and formatting interventions, and calibrates iframe positions. Explicit target formatting improved the original deployments from 0/40 to 35/40 against a contemporaneous baseline, although separation curves were not uniformly monotonic across deployments. Independent collectors replicate the iframe reachability null; calibrated vision trials show no consistent depth gradient. The original variance decomposition remains a confounded descriptive audit. Observation boundaries belong to versioned stacks, and available evidence still requires reliable delimitation, localization, and action execution.

1 INTRODUCTION

Web agents complete tasks by observing pages and selecting actions (Deng et al., 2023; Zhou et al., 2024; He et al., 2024). Their observations may serialize the Document Object Model (DOM), expose the accessibility tree, capture screenshots, or traverse browser protocols. Final task success alone does not distinguish evidence omitted by an observation pipeline from failures to interpret, locate, or act on available evidence.

Cross-origin iframes separate browsing contexts, shadow roots encapsulate component subtrees, and canvas exposes pixels without corresponding textual DOM content. These structures make evidence availability consequential. Broader observation privileges can recover embedded content but also create security exposure under prompt injection (Roesner & Kohlbrenner, 2026). Capability to cross a boundary must therefore be distinguished from permission to do so.

Prior work studies observation modality, visual grounding, stage-wise GUI failures, and controlled visual perturbations (Koh et al., 2024; Zheng et al., 2024; Ji et al., 2026; Yu et al., 2026). Our benchmark isolates a complementary quantity: whether task-relevant evidence enters a versioned observation pipeline at all, measured independently of subsequent interpretation and action.

We contribute (1) a version-pinned benchmark varying iframe depth and origin, shadow containment, and rendering while holding task semantics fixed; (2) paired reachability and task-success scores, with archived-response coding of post-capture failures; and (3) controlled follow-ups across four vision deployments testing separation, familiar layouts, visible boundaries, explicit target formatting, and calibrated iframe positions. Independent implementation probes test whether the iframe result

survives a change of collector. The aim is diagnostic: to locate observation loss and distinguish it from failures after capture.

2 RELATED WORK

World of Bits exposed both DOM and rendered pixels (Shi et al., 2017); Pix2Act learns screenshot-to-action interaction (Shaw et al., 2023). WebArena and WebArena Verified provide execution-based evaluation (Zhou et al., 2024; El Hattami et al., 2025), while VisualWebArena tests multimodal tasks (Koh et al., 2024). BrowserGym standardizes observation/action spaces and experiment management across benchmarks (Le Sellier De Chezelles et al., 2025). We complement these task-level evaluations with an independent evidence-availability score attached to a versioned collector.

SeeClick links screenshot grounding improvements to downstream GUI success (Cheng et al., 2024). SeeAct identifies grounding as a bottleneck and finds hybrid HTML–visual grounding more effective than its visual-prompting alternatives (Zheng et al., 2024). FineState-Bench separately diagnoses localization, interaction, and exact target-state attainment (Ji et al., 2026). Our reachability score addresses an earlier question—whether the evidence-bearing content entered the observation—without requiring a successful model interpretation or action. It complements, rather than replaces, such stage-wise diagnostics.

StressWeb and WebForge study controlled perturbations (Bai et al., 2026; Yuan et al., 2026); ST-WebAgentBench includes a canvas challenge (Levy et al., 2026). VAF holds semantics fixed while varying visual attributes and measures changes in agent choice (Yu et al., 2026). Our separation and boundary studies instead measure target delimitation and submission alongside reachability. Set-of-Mark overlays labeled regions to support visual grounding (Yang et al., 2023); our answer-format instruction tests a distinct intervention, without supplying an oracle region or testing a DOM cross-check.

InjecAgent and AgentDojo evaluate indirect prompt injection through external content and tool outputs (Zhan et al., 2024; Debenedetti et al., 2024). Together with browser-specific isolation concerns (Roesner & Kohlbrenner, 2026), they motivate separating observational capability from authorized access. Our self-authored fixtures measure capability, not adversarial robustness.

3 BENCHMARK AND METHODS

3.1 TASKS AND STRUCTURAL AXES

The benchmark contains two deterministic tasks: reading a known value and copying a known value into a target field. Each task first passes a level-0 control consisting of a flat, single-frame, shadow-free, canvas-free page. Generated variants preserve the task instruction, correct answer and marker tokens, and external checker while changing one of four ordered axes:

1. same-origin iframe depth 0–3;
2. cross-origin iframe depth 0–3;
3. shadow containment (none, open, closed, nested open–open); and
4. rendering target (ordinary DOM or canvas).

The shared level-0 condition yields 11 unique structural conditions. Pages and checkers were self-authored and deterministic; no live or authenticated service was included.

Fixed task content. Across conditions the instruction string, correct answer token, source label, and task-specific marker token are identical. The complete visible strings and layout are not byte- or pixel-identical: DOM pages place the source and prefixed marker in separate elements, whereas canvas draws the source label, value, separator, and unprefixed marker as one line. The experiment therefore holds task semantics and the two load-bearing tokens fixed while changing both representation and the boundary cues that representation provides.

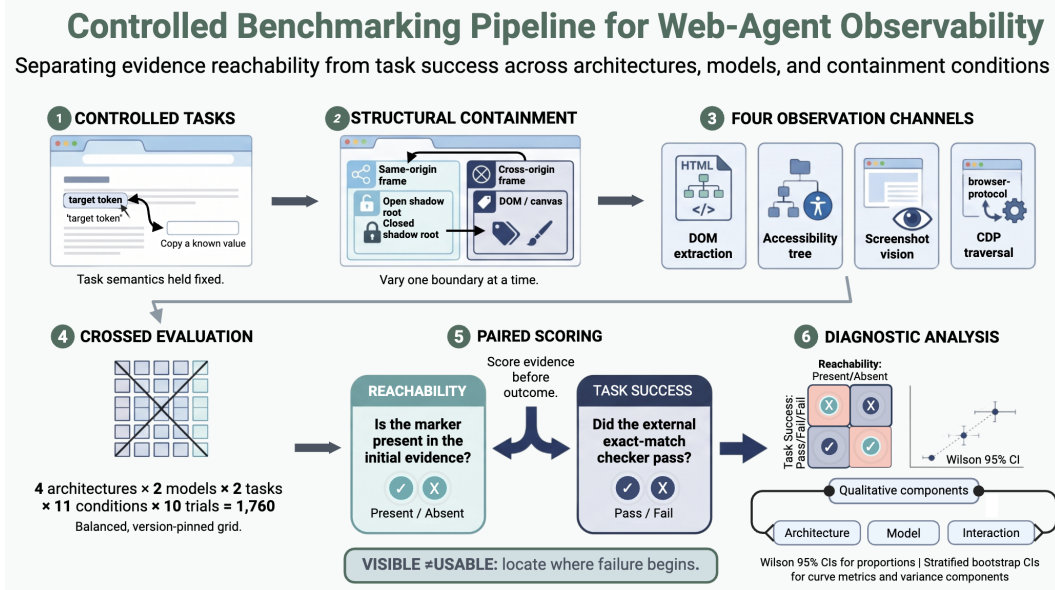


Figure 1: Controlled benchmarking pipeline for web-agent observability. Fixed task semantics are crossed with structural-containment conditions and four observation channels; reachability and exact-match task success are scored separately before diagnostic analysis.

Separation and revision studies. The original six-layout vision follow-up held tokens, typeface, task, and checker fixed while varying target–marker separation, with ten repeats per task and deployment ($N = 240$). The revision adds 240 trials on `gpt-5.4` and `gpt-5-mini` using those same fixtures. Three familiar shells—tables, cards, and forms—cross text-box edge gaps of 8/40/96 pixels with DOM distances of zero/four parent edges ($N = 1,440$). Visually identical DOM variants serve as negative controls; a further 240 trials use visibly distinct labeled components. A paired inline-canvas study tests explicit target formatting ($N = 160$). Finally, 560 calibrated iframe trials cross four deployments, two tasks, seven conditions, and ten repeats, with positions randomized by task/repeat and matched across conditions. All new arms retain eight actions and 90 seconds. Appendix F specifies geometry, prompts, scoring, and coordinate checks.

3.2 OBSERVATION ARCHITECTURES AND CROSSED MODELS

We evaluate four architecture classes defined by their page representation: *DOM extraction* serializes page structure through an automation framework; *accessibility tree* consumes the browser-computed accessibility representation; *vision* receives screenshots and emits coordinate actions; and *Chrome DevTools Protocol (CDP) traversal* consumes browser-level DOM snapshots and frame state. One version-pinned implementation instantiates each class (Appendix A). The labels therefore denote the tested pipelines rather than universal properties of every product using the same modality.

Each architecture is crossed with the same two hosted base models in the primary grid. Every task–model–condition cell contains ten valid trials, giving $4 \times 2 \times 2 \times 11 \times 10 = 1,760$ outcomes. Before the sweeps, a 48-trial control gate required all architecture–model pairs to pass both control tasks three times. In response to the small original roster, a separately versioned vision-only extension adds `foundry/gpt-5.4` and `foundry/gpt-5-mini` on the same two tasks, 11 conditions, and ten-trial allocation (440 additional valid outcomes). Run manifests preserve framework, browser, model deployment, timestamp, task revision, prompt version, and source hashes.

3.3 OUTCOMES AND ANALYSIS

Reachability indicates whether the task-specific marker appears in the architecture’s initial task-page evidence; it is rescored from immutable archives without another model call. For the three structure-based architectures this is a text-match test against the archived initial-page serialization.

Table 1: Overall outcomes and recorded resource use across all 11 conditions ($N = 440$ per architecture). Tokens are mean input plus output tokens; provider-dollar cost was not present in archived responses. Success is scored by exact match; see Section 4.3 for a relaxed-criterion sensitivity analysis of the vision canvas cells.

Architecture	Reachability	Success	Actions/run	Tokens/run
DOM extraction	400/440	400/440	2.46	14,839
Accessibility tree	400/440	398/440	2.03	14,927
Vision	440/440	357/440	4.16	8,645
CDP traversal	400/440	399/440	2.71	8,765

Vision likewise uses no model call: a separate privileged Playwright probe searches every page frame for a benchmark selector (the marker element, target canvas, or shadow host), obtains its bounding box, and scores the trial reachable when at least one nonzero box intersects the 1280×900 captured viewport. For the canvas condition, the operative selector is `#target-canvas`. The probe does not perform optical character recognition or pixel-template matching, and none of its metadata enters the model request. Reachability therefore measures presence of the evidence-bearing fixture region in the observation, and is scored independently of whether the model transcribed, delimited, or acted on it.

Task success is determined by an external state checker using exact match against the correct answer. Strict equality was fixed before the sweeps; Section 4.3 reports a post hoc sensitivity analysis under a relaxed criterion, because the observed canvas failures were compositional rather than substitutional.

For canvas sensitivity analyses we score the emitted answer (read) or submitted field value (copy) under three content criteria: exact match after outer-whitespace stripping; target present as a substring; and target present with the task-specific marker absent. The third permits explanatory wrapper text but rejects the observed target-marker concatenation.

We additionally record actions, input tokens, output tokens, elapsed time, and four joint outcomes formed by crossing reachability with success. Success conditional on unreachable evidence is reported only as an operational confabulation proxy, not as a claim about model intent.

Failure coding. For every failed vision trial we additionally code, from the archived model response and action log, where in the pipeline the failure occurred: extraction (the correct answer token is absent or altered in the response), delimitation (the correct token is present but bounded incorrectly, so the emitted string is a superset or subset of the answer), localization (the response is correct but coordinate actions target the wrong element), action execution (correct content and target, but the emitted action sequence does not reach the checked state), and abstention. Coding operates on preserved archives and requires no additional model calls.

Proportions use two-sided Wilson 95% confidence intervals (CIs). Reachability estimates and Figure 2A pool the two tasks and two primary models within an architecture ($N = 40$ per condition), as reachability is scored without a model call. Figure 2B combines the primary and extension deployments on the matched original-layout vision-canvas cells while pooling tasks ($N = 20$ per deployment); Appendix B reports the underlying task-model cells ($N = 10$). The four-model vision results are otherwise kept deployment-resolved, and the extension models do not enter architecture comparisons because they were not run on the three structure-based stacks. Curves use normalized trapezoidal area under the curve (AUC), total level-0-to-final drop, largest adjacent drop, and a preregistered cliff indicator. The registered variance decomposition is retained only as a confounded descriptive audit in Appendix G. Full definitions appear in Appendix C.

4 RESULTS

All four sweeps were complete and balanced at 440 valid outcomes per architecture. The control gate passed 48/48 trials. A provider-free rescore of the DOM and accessibility-tree archives under the final initial-page policy changed no trial labels.

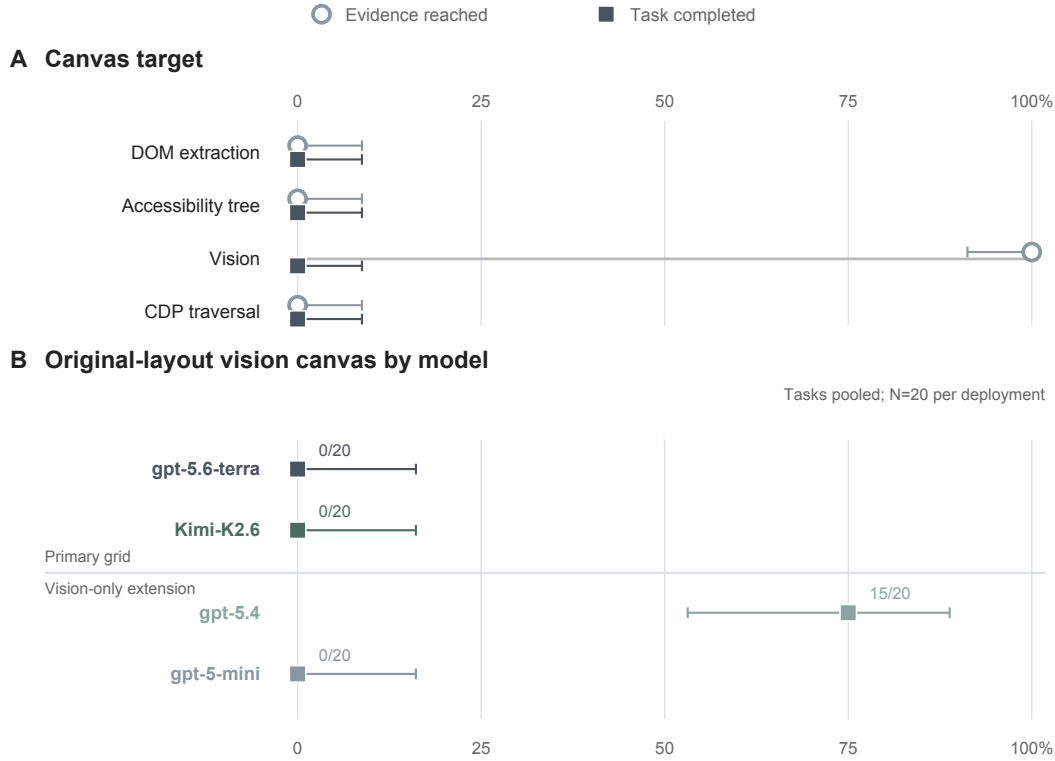


Figure 2: **Evidence availability and task completion separate under visual observation.** (A) Primary-grid canvas estimates pool two tasks and two models ($N = 40$). Structure-based observations neither reached nor completed the task; vision reached 40/40 targets but completed 0/40 under exact match. (B) The matched original-layout vision-canvas cells combine the two primary and two extension deployments while retaining deployment-level estimates ($N = 20$ each): `gpt-5.4` completed 15/20, whereas the other three completed 0/20. Whiskers are Wilson 95% confidence intervals; Appendix B gives the $N = 10$ task-model cells and the full four-deployment vision grid.

4.1 REACHABILITY WAS FLAT EXCEPT AT THE RENDERING BOUNDARY

Every architecture exposed the target marker in 40/40 trials for every non-canvas condition (Wilson 95% CI 91.2–100%). Thus neither iframe origin or depth nor open, closed, or nested shadow containment caused measured reachability loss in the tested implementations. Because structure-based reachability is scored by text match and vision reachability by a privileged Playwright probe, neither score requires or depends on a model call. Pooling reachability over models is therefore appropriate by construction. A targeted replication then tested the iframe cells using Selenium recursive frame switching as a second DOM implementation and Playwright’s public per-frame Accessible Rich Internet Applications (ARIA) snapshot as a second accessibility implementation. Each reached 16/16 task–origin–depth cells. The null therefore held in both tested DOM implementations and both tested accessibility implementations, rather than only in browser-use and one raw-CDP accessibility collector. This result contradicts the cross-origin and closed-shadow cliffs predicted by the mechanistic account that motivated the design.

Canvas produced the sole reachability separation (Figure 2A). DOM extraction, accessibility tree, and CDP traversal each fell from 40/40 on the ordinary DOM target to 0/40 on canvas (0%; 95% CI 0–8.8%). Vision remained at 40/40 because the target pixels remained in the screenshot. Under the cliff rule, every task–model curve in the three structure-based architectures contained a complete canvas cliff; vision contained none.

4.2 TASK SUCCESS SEPARATED VISIBILITY FROM EXECUTION

Success closely tracked reachability for structure-based observations. Excluding canvas, DOM extraction completed 400/400 trials, accessibility tree 398/400 (99.5%), and CDP traversal 399/400 (99.8%); all three then completed 0/40 canvas trials. Among the 40 unreachable trials for each of these architectures, none succeeded, giving a confabulation-proxy estimate of 0% (95% CI 0–8.8%). Vision had no unreachable trials, so its conditional proxy is undefined rather than zero.

Vision exposed the complementary failure mode (Figure 2B): the original deployments reached all 40 canvas targets but completed none, while completing all 120 shadow-containment trials. The original iframe success pattern was confounded by screenshot scaling and fixed absolute positions; Appendix F preserves the full audit. The earlier 440-trial vision extension showed original-layout heterogeneity: `gpt-5.4` completed 15/20 canvas trials and `gpt-5-mini` 0/20. These fixed-position results motivate the separation extension and calibrated rerun below, rather than a depth claim.

4.3 CANVAS FAILURE WAS A DELIMITATION FAILURE, NOT BLINDNESS

Coding the 40 failed canvas trials from the two original deployments against the archived model responses and action logs, both deployments failed all 20 trials each. Across all 40, 40/40 misreported the exact target value, 0/40 reported it correctly but mislocalized the `Destination` field, 0/40 reached correct extraction and localization but failed during action execution, and 0/40 refused or abstained. Every response appended the visible control marker to the true token; in all 20 copy trials, the checker recorded that composite string in the `Destination` field. Within that preregistered roster, failure was entirely extraction—specifically, target delimitation—rather than grounding or action execution. The later `gpt-5.4` successes show that the behavior is not model-universal.

The original failure was compositional rather than substitutional: the correct token was recovered from pixels in every coded trial and was then over-included rather than replaced or hallucinated. The follow-up below identifies the operative factor more narrowly: close visual layout makes the answer boundary ambiguous, while structural serialization can supply a boundary when the tokens occupy distinct elements. Canvas is one way to remove those structural cues, not a sufficient cause of failure by itself.

Checker sensitivity. Because all 40 responses contained the target token as a substring, the canvas success rate depends jointly on rendering and on scoring strictness. We therefore report three criteria: the preregistered exact match (0/40), a permissive substring check (40/40), and an intermediate check allowing wrapper text but requiring that the target appear and the task-specific marker be absent (0/40). The last criterion rejects precisely the observed boundary error without requiring bare-token output. Thus the claim that these trials failed to delimit target from marker does not depend on choosing exact match over substring; substring instead serves as evidence that the target glyphs were read.

The original deployments show a separation dose–response. For the original two deployments in the controlled follow-up, exact-match and target-without-marker criteria agreed at every level and increased monotonically from 0/40 for the inline bar, to 0/40 for two spaces, 10/40 for a 40-pixel gap, 31/40 for separate lines, and 38/40 for opposite corners. The corresponding Wilson 95% intervals were 0–8.8%, 0–8.8%, 14.2–40.2%, 62.5–87.7%, and 83.5–98.6%. This dose–response is specific to the original pooled deployments. The same-element DOM inline-bar control also scored 0/40 under exact and target-without-marker criteria while retaining the target substring in 40/40. Thus the original failure is not uniquely caused by canvas rendering; it is a layout-driven delimitation failure that canvas serialization cannot repair with element boundaries. Under the permissive substring criterion, the five canvas levels scored 37/40, 35/40, 29/40, 37/40, and 38/40; the nonmonotonic values reflect complete timeout/failure transcripts concentrated in Kimi copy trials rather than target–marker boundary acceptance.

4.4 GENERALIZATION AND A TARGETED INTERVENTION

The separation curve depends on the deployment and task. The two added deployments were tested on the same six-layout follow-up fixtures as the original deployments. Across inline bar, two

Separation curves remain deployment- and task-dependent

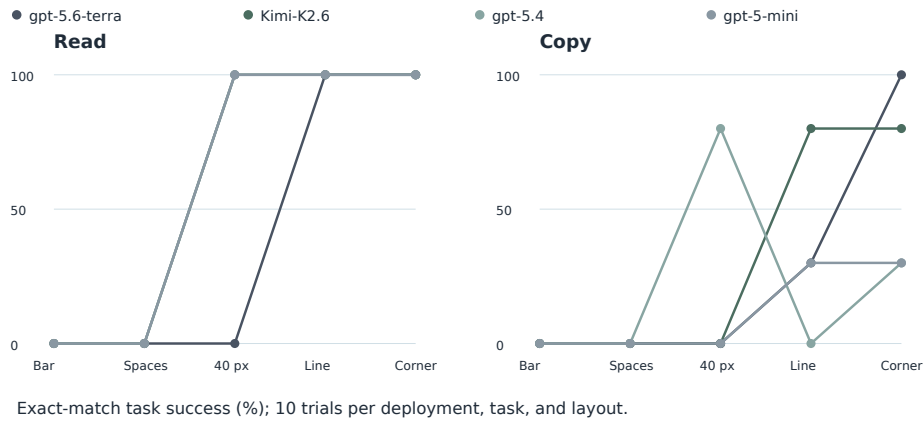


Figure 3: **Task success depends on separation, task, and deployment.** Exact-match success is shown separately for reading and copying, with ten trials per deployment–task–layout cell. The original two deployments and the two added deployments use the same follow-up fixtures and budgets in separate run windows. The full three-criterion estimates and Wilson intervals, including the DOM inline control, are in the artifact.

spaces, 40-pixel gap, separate lines, and opposite corners, respectively, gpt-5.4 scored 0/20, 0/20, 18/20, 10/20, 13/20. gpt-5-mini scored 0/20, 0/20, 10/20, 13/20, 13/20. Figure 3 retains the task split; the full three-criterion cell estimates are in the artifact. The original-layout 15/20 result for gpt-5.4 is therefore evidence about that fixture, not a deployment-wide separation threshold.

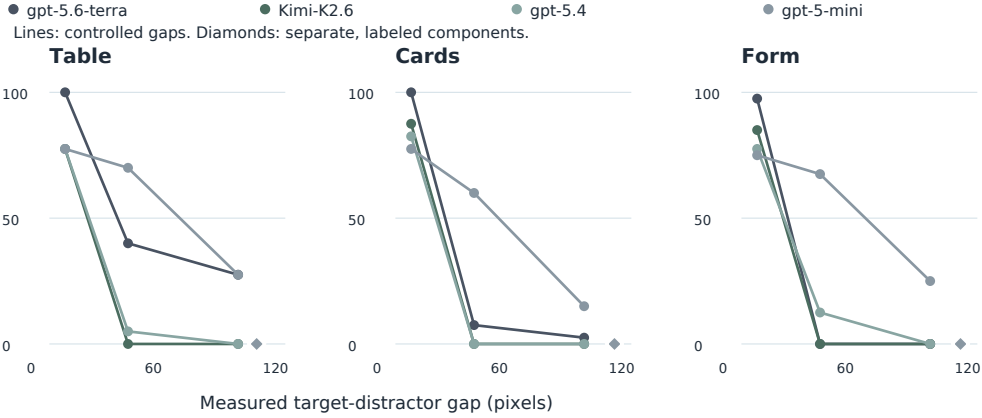
Measured distance in familiar interface shells. Figure 4A reports delimitation error against measured pixel distance. Across the 360 trials per deployment, exact task success was gpt-5.6-terra 108/360, Kimi-K2.6 116/360, gpt-5.4 115/360, gpt-5-mini 41/360. Delimitation is scored from the final read answer or last attempted typed content; correct content with a failed submission is kept separate. The full estimates retain layout, task, deployment, pixel distance, and DOM separation. Because the DOM variants have the same visual appearance, their contrast is a negative control rather than a test of structural access by vision. These controlled shells broaden layout coverage but do not establish real-site generalization.

Increasing the gap from 8 to 96 pixels reduced delimitation errors by 89.2, 83.3, 79.2, and 54.2 percentage points for gpt-5.6-terra, Kimi-K2.6, gpt-5.4, and gpt-5-mini, respectively; each paired 95% bootstrap interval excluded zero. Hidden-DOM contrasts ranged from -2.2 to +3.3 points, with all intervals including zero. Full intervals are in the artifact. Lower delimitation error alone does not imply higher task success: missing answers and failed submissions remain separate outcomes.

The natural-boundary controls use separate table columns, neighboring cards, and labeled form fields ($N = 240$). Across 60 trials per deployment, exact success was gpt-5.6-terra 59/60, Kimi-K2.6 45/60, gpt-5.4 47/60, gpt-5-mini 33/60. No delimitation errors were observed in these 240 trials. Figure 4A places these controls at their measured mean pixel distance. They change component boundaries and labels together and therefore support a compound layout comparison, not an isolated causal effect of DOM distance.

Explicit target formatting. Figure 4B compares contemporaneous paired baseline and format-prompt trials. gpt-5.6-terra changed from 0/20 to 20/20 (+100 percentage points; paired bootstrap 95% CI +100 to +100); Kimi-K2.6 changed from 0/20 to 15/20 (+75 percentage points; paired bootstrap 95% CI +60 to +90); gpt-5.4 changed from 1/20 to 11/20 (+50 percentage points; paired bootstrap 95% CI +35 to +65); gpt-5-mini changed from 0/20 to 11/20 (+55 percentage points; paired bootstrap 95% CI +50 to +65). This tests an answer-format instruction, not an oracle

A Delimitation error (%) across familiar layouts



B Explicit format prompt: inline-canvas task success

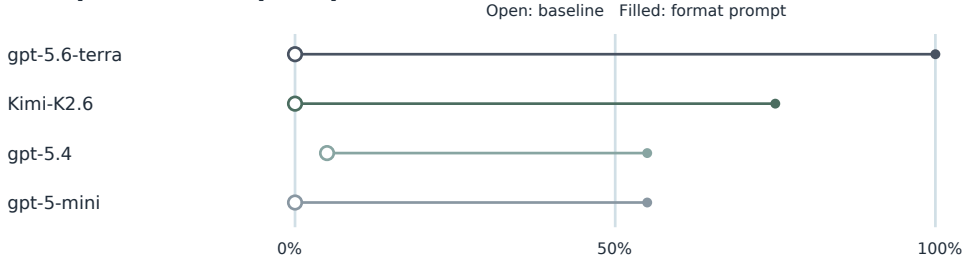


Figure 4: **Generalization by measured distance and a targeted prompt intervention.** A: delimitation-error rates in tables, cards, and forms, pooling tasks and the two visually matched DOM boundaries ($N = 40$ per deployment–layout–gap point). Diamonds show separate labeled components at their measured mean distance ($N = 20$); all four deployments overlap at zero observed error. B: exact task success for inline-canvas baseline (open) and explicit-format prompts (filled), with $N = 20$ per deployment and arm. Cell-level Wilson intervals and paired-effect intervals are provided in the artifact.

region or DOM cross-check. The comparison uses its own matched baseline because the calibrated inline fixture differs in viewport and canvas width from the original 0/40 result.

Iframe depth after calibration and position matching. The calibrated rerun obtained all 560 allocated outcomes. Pooled over the seven conditions, read/copy success was gpt-5.6-terra 70/70 / 70/70, Kimi-K2.6 70/70 / 56/70, gpt-5.4 70/70 / 44/70, gpt-5-mini 70/70 / 10/70. Reading was 10/10 at every depth/origin cell for every deployment. Copying was 10/10 throughout for gpt-5.6-terra; the other deployments varied non-monotonically across depths. There is no consistent depth gradient in these outcomes. Appendix H retains all task–deployment–depth cells; paired comparisons against depth zero are in the artifact. Since input screenshots are position-matched across depths, residual differences cannot be attributed to the old fixed screenshot-scale or absolute-position artifact. Ten paired repeats per cell limit sensitivity; a small or nonsignificant contrast is not evidence of equivalence. The original variance decomposition remains only a descriptive audit in Appendix G.

4.5 OBSERVATION CHANNELS HAD DIFFERENT RESOURCE PROFILES

Resource use did not follow a simple “vision is more expensive” pattern. Vision took the most actions but used fewer tokens overall and per action (2,076 tokens/action) than DOM extraction (6,034), accessibility tree (7,338), or CDP traversal (3,230). Per-action figures are means of per-trial ratios and therefore differ slightly from the ratio of the pooled means in Table 1. Accessibility-tree tokens per

action rose most sharply with iframe depth, consistent with increasingly large serialized tree contexts. Appendix E shows the pooled iframe-depth curve; the artifact includes the complete per-condition cost curves. Because archived responses omitted billing amounts, we report actions and tokens rather than imputing dollar cost.

5 DISCUSSION, LIMITATIONS, AND FUTURE WORK

The results show why evidence availability and task completion must be evaluated separately. Iframe and shadow containment preserved evidence in the tested stacks, whereas canvas exposed a gap between structural observations and screenshots. Within the primary grid, each architecture is represented by one versioned implementation; independent DOM and accessibility collectors strengthen the iframe finding without establishing a universal property of either modality. Once evidence was available to vision, delimitation and subsequent action remained distinct sources of failure.

This distinction makes answer boundaries a practical design concern. Explicit target formatting improved the original deployments from 0/40 to 35/40 against a contemporaneous baseline, while separate labeled components yielded no observed delimitation errors. The latter comparison changed labels, geometry, and boundaries together, so it cannot identify their individual effects. Likewise, the original canvas fixture coupled rendering with close layout. The separation sweep, DOM control, and scoring sensitivity analysis clarify this mechanism, but deployment- and task-dependent responses do not establish a universal separation threshold. Correctly recovered content also did not guarantee successful submission.

The scope of these findings is limited by the two synthetic tasks, which isolate mechanisms without establishing performance on long-horizon websites or combined structural barriers. Model coverage imposes a further constraint: the added deployments were tested only through vision. Broader architecture comparisons therefore require the complete model roster crossed with all observation stacks. For depth effects, calibrated trials remove the original scale and position confounds, but ten paired repeats per cell cannot establish equivalence. Larger paired samples are needed; resampling cannot remove the original fixture confounds or make the registered variance shares causal.

The diagnostic measures also require careful interpretation. Reachability establishes evidence availability rather than comprehension, and failure labels infer stages from archived behavior rather than latent states. Blinded coding agreed on all 40 cases, but the single observed category leaves ordinary Cohen’s κ undefined; Appendix D reports the sensitivity statistic. Similarly, the zero-success confabulation proxy retains an 8.8% upper Wilson bound and cannot rule out rare unsupported success.

These limits motivate evaluating interventions against both content accuracy and completed actions. Set-of-Mark visual prompts and SeeAct’s hybrid grounding suggest region prompts and structural cross-checks (Yang et al., 2023; Zheng et al., 2024), which remain untested here. Vision’s higher action count but lower token use further motivates selective hybrid observation. Such evaluations should measure when an additional observation improves completion and at what resource cost; provider-dollar cost and machine-level energy remain unmeasured.

Extending this agenda to real sites also requires separating capability from permission. Automation can cross boundaries that constrain page scripts, so broader reach alone does not justify broader access. Future studies should combine structural barriers on permission-respecting sites and compare restricted observation policies under adversarial instructions and sensitive decoys, testing whether gains in completion persist within authorized access boundaries.

6 CONCLUSION

Iframe and shadow containment caused no reachability loss in the tested stacks, while canvas separated screenshot from structural evidence. The follow-ups reveal deployment- and task-dependent separation curves and a practical benefit from explicit target formatting. Calibrated iframe cells show perfect reading and no consistent copy-success gradient with depth, without establishing equivalence. Evaluations should attach claims to versioned stacks and distinguish missing evidence, delimitation, localization, and action execution.

AI USE STATEMENT

Generative AI assistance (OpenAI Codex) was used to implement benchmark runners, synthetic fixtures, scoring, analysis utilities, and polish figures; coordinate experiment execution; refine follow-up experimental designs; interpret archived results; and revise and format manuscript text. The historical failure-coding audit also used an independent blinded LLM coding pass, as described in Appendix D. Validation included automated scoring tests, fixture geometry and click/submit checks, archived-response inspection, paired-image hash audits, and visual inspection of the generated figures and PDF. Benchmark-model outputs are retained as experimental observations rather than treated as ground truth. The authors remain responsible for the final text, code, data, claims, citations, and artifacts.

ETHICS STATEMENT

All URLs and assets were self-authored synthetic fixtures. The motivating healthcare portal was never accessed or included; no personal data, third-party accounts, human participants, or crowdsourced labor were used. Broader cross-origin reach may enable data exfiltration under prompt injection, so capability should be separated from permission and paired with access controls, provenance logging, and evaluation of restricted versus unrestricted modes before deployment.

REPRODUCIBILITY STATEMENT

Section 3 defines the tasks and outcomes. Appendices A and F document the versioned stacks, fixture geometry, prompts, run allocation, and coordinate checks; Appendices C and H give statistical definitions and calibrated cells. The accompanying revision tables and manifests record all 2,640 new valid trials and their provenance. The original anonymous repository is linked in Appendix J; original and revision run windows remain distinct.

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SUPPLEMENTARY MATERIAL

A IMPLEMENTATION AND RUN PROVENANCE

Table 2: Version-pinned implementations used in the completed grid. Model deployment identifiers are the aliases recorded in the manifests.

Class	Implementation	Prompt/version
DOM extraction	browser-use 0.13.7 at commit f0aa3a8; Chromium 1228	dom-no-vision-v2
Accessibility tree	Playwright 1.61.0; Chromium 1228; CDP accessibility snapshot	ax-full-action-v3
Vision	Playwright 1.61.0 screenshot and coordinate-action runner; Chromium 1228	vision-coordinate-v1
CDP traversal	Playwright 1.61.0 CDP DOM-snapshot runner; Chromium 1228	cdp-snapshot-v1

The two primary Azure AI Foundry deployments were recorded as `foundry/gpt-5.6-terra` and `foundry/Kimi-K2.6`. The vision-only extension added `foundry/gpt-5.4` and `foundry/gpt-5-mini`; its 440/440 valid outcomes are hash-bound to the primary vision-resolved specification. Primary-grid execution ran from 2026-08-30 17:55:16 to 2026-08-31 02:17:28 UTC, and the provider-free reconciliation manifest was written at 02:33:59 UTC. The 2026-09-04 follow-ups have separate immutable windows: the implementation probes ran 14:38:00–14:38:04 UTC, the 240-trial separation sweep 14:51:51–15:10:19 UTC, and the 440-trial vision extension 15:28:33–16:02:50 UTC. These dates are reported separately rather than retroactively changing the primary receipts. The DOM, accessibility, vision, and CDP receipts each contain task revisions, condition and model rosters, valid-trial counts, source and result hashes, exclusions, and completion verdicts. The final primary control gate used the same four architecture classes and two models and passed all 48 required trials.

For the iframe replication, Selenium 4.48.0 recursively switched into each frame and inspected that frame’s own `page_source`; Playwright 1.61.0 independently obtained a public `aria_snapshot` from each frame. Neither path reused browser-use’s serialization or the primary accessibility runner’s raw `Accessibility.getFullAXTree` call. Both reached all 16 task-origin-depth cells.

Revision-run provenance. The new separation, generalization, visible-boundary, intervention, and calibrated-iframe arms contain 240, 1,440, 240, 160, and 560 valid trials, respectively (2,640 total). Recorded new-trial timestamps span 2026-09-15T18:31:00.151404Z to 2026-09-16T00:29:10.284138Z. Each arm retains its own frozen fixture source, matrix, geometry, model identifiers, and result hash. Interrupted runs inherit completed outcomes unchanged and resume only unfinished or infrastructure-failed cells. The attached paper’s text matches the archived editing source across all 13 pages. The revision package supplies per-trial measurements and cell estimates; raw local archives retain screenshots, actions, prompts, and responses. These new artifacts are separate from the original public artifact snapshot.

B MODEL-RESOLVED TASK-SUCCESS CELLS

The headline primary-grid results remain pooled where the split is uninformative. Table 3 confirms that both primary deployments were at or near ceiling for every non-canvas structure-based architecture. Figure 2B combines all four deployments on the matched original-layout vision-canvas cells; Table 4 reports the complete four-deployment vision grid. The extension models were not run through the three structure-based agent stacks and therefore do not enter an architecture-by-model decomposition.

Table 3: Non-canvas task success for the three structure-based architectures, shown by model deployment. Each model column contains $N = 200$ trials; the pooled column contains $N = 400$.

Architecture	gpt-5.6-terra	Kimi-K2.6	Pooled
DOM extraction	200/200	200/200	400/400
Accessibility tree	200/200	198/200	398/400
CDP traversal	200/200	199/200	399/400

Table 4: Complete four-deployment vision iframe task-success grid. Entries are successes out of ten trials. S denotes same-origin and X cross-origin; depth 0 is the shared level-0 control and is repeated across the two blocks for comparison. The bottom four rows are the vision-only extension.

Model deployment	Task	S0	S1	S2	S3	X0	X1	X2	X3
foundry/gpt-5.6-terra	Copy	10/10	0/10	10/10	10/10	10/10	0/10	10/10	9/10
foundry/gpt-5.6-terra	Read	10/10	10/10	10/10	0/10	10/10	10/10	10/10	0/10
foundry/Kimi-K2.6	Copy	10/10	10/10	10/10	10/10	10/10	10/10	10/10	8/10
foundry/Kimi-K2.6	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
foundry/gpt-5.4	Copy	10/10	2/10	0/10	7/10	10/10	0/10	7/10	2/10
foundry/gpt-5.4	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
foundry/gpt-5-mini	Copy	10/10	10/10	9/10	7/10	10/10	9/10	9/10	8/10
foundry/gpt-5-mini	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10

C STATISTICAL DEFINITIONS

For a curve with ordered rates $y_0, \dots, y_L$, normalized AUC is the equally spaced trapezoidal area divided by L ; total drop is $y_0 - y_L$; and the largest single-level drop is $\max_\ell (y_{\ell-1} - y_\ell)$. The cliff ratio is the largest adjacent drop divided by total drop. Curves with total drop below 0.20 are labeled `no_material_net_drop`; remaining curves are labeled `cliff` when the ratio is at least 2/3 and `slope` otherwise. Threshold sensitivity at 0.50 and 0.75 was also computed.

Wilson intervals use the two-sided 95% score interval. Curve-metric and variance-component intervals use a deterministic stratified nonparametric bootstrap: trials are resampled with replacement within each task-model-condition cell, and the complete statistic is recomputed for 2,000 resamples. Where a contrast is deterministic within cells, as for the canvas reachability split, the resulting interval is degenerate and carries no information about sampling variability. Curve shape for the variance decomposition is represented by level-0-minus-level rates at every nonbaseline axis level, averaged over the two tasks. A balanced two-way sum of squares partitions variation into architecture, model, and interaction shares, normalized by their sum.

New-study rates use Wilson intervals. Paired-effect intervals use 10,000 bootstrap samples of complete randomized-position repeat blocks within task strata, preserving paired conditions and the layout/gap observations sharing each position. Degenerate paired bootstrap intervals reflect no observed variation, not certainty about future trials; cell Wilson intervals remain informative at ceiling. These exploratory comparisons use no equivalence margin or multiplicity adjustment.

D FAILURE CODING SCHEME

Each failed vision trial is assigned exactly one category, applied in order, from the archived response text and action log:

1. **Extraction—substitution:** the correct answer token does not appear in the response.
2. **Extraction—delimitation:** the correct answer token appears, but the emitted answer is a strict superset or subset of it (for example, an adjacent visible string is appended).
3. **Localization:** the emitted answer is exactly correct, but coordinate actions targeted an element other than the specified field.

4. **Action execution:** content and target are both correct, but the emitted action sequence did not produce the checked state.
5. **Abstention:** the model declined, reported that it could not find the target, or emitted no answer.

Categories are mutually exclusive and jointly exhaustive over failed trials. Coding operates only on preserved archives and involves no additional provider calls, so it inherits the provider-free reproducibility of the rest of the analysis. Initial labels were assigned deterministically by exact-string and substring comparison against the known target and marker, then hand-checked against all 40 archived response and action records. A second coding agent independently received a randomized packet with opaque case identifiers and was blinded to model, condition, repeat, run identifier, first-coder labels, and aggregate results. It assigned all 40 cases to extraction failure, matching the first coder in 40/40 cases. Ordinary Cohen’s κ is undefined because both marginal label distributions have zero variance (expected chance agreement is 1.0); raw agreement is 100%, and prevalence-adjusted bias-adjusted $\kappa = 1.00$ is reported only as a sensitivity statistic.

E RECORDED RESOURCE CURVES

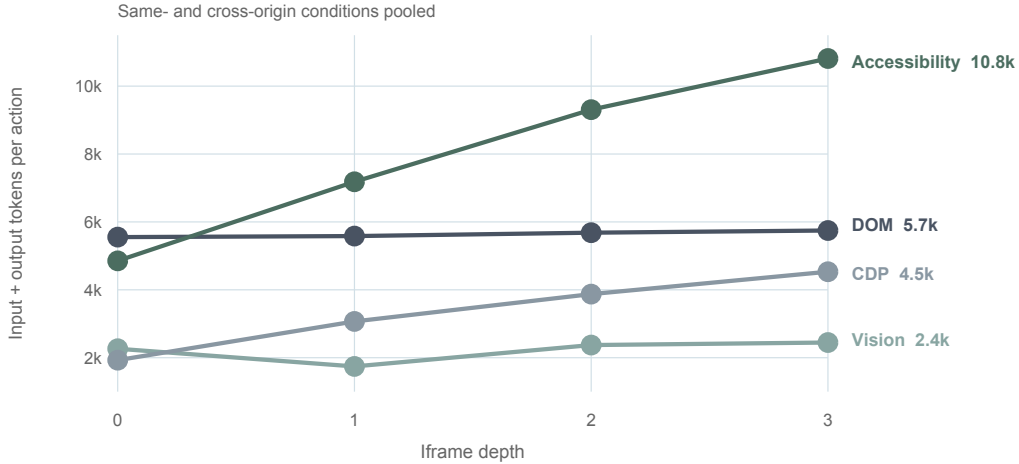


Figure 5: **Serialized context grows differently across observation channels.** Lines show mean input-plus-output tokens per recorded action across iframe depth, pooling same- and cross-origin conditions, tasks, and models. The shared depth-0 control has $N = 40$ per architecture; depths 1–3 have $N = 80$ each. Overall actions and tokens per run are reported in Table 1; complete per-condition estimates are included in the artifact.

F FIXTURE AND REVISION-STUDY DETAILS

Fixture geometry. Because the canvas result below turns on visual separation between adjacent strings, we state the fixture layout explicitly. Each task page displays the target value together with a visible task-specific control marker—`CONTROL_READ_VALUE_R2` for reading and `CONTROL_COPY_VALUE_R5` for copying—which is present in every condition and is not part of the correct answer. In DOM conditions, the source label and value occupy one block-level paragraph, while the marker appears in a separate later paragraph in smaller text prefixed by “Target marker:”; the copy form intervenes between those paragraphs in the copy task. In the canvas condition, a single `fillText` call draws the complete string at (20, 80) in a 900×160 canvas using 24-pixel sans-serif type, with the target and marker on the same line separated only by one space, a vertical bar, and one space (|). The copy task requires writing the target value—and only the target value—into a field labeled `Destination`.

Visual-separation follow-up. To separate rendering modality from layout, a new fixture family holds task, typeface, tokens, and checker fixed while placing the target and marker (i) inline with a bar, (ii) inline with two spaces, (iii) inline with a 40-pixel gap, (iv) on separate lines, or (v) at opposite canvas corners. A sixth control renders the inline-bar string as ordinary DOM text inside one element. Both original vision deployments receive ten trials per task and layout ($N = 240$). The follow-up captures at 1092×768 , matching the high-detail image dimensions delivered to the model so screenshot and action coordinates share one space, and uses a fixed eight-action, 90-second budget. As in the primary sweep, complete timeout transcripts count as task failures; only infrastructure errors are retried and excluded. All 240 allocated outcomes were obtained with no infrastructure exclusions.

Deployment extension, familiar layouts, and intervention. A separately versioned 240-trial run extends the identical six-layout separation sweep to `gpt-5.4` and `gpt-5-mini`. We also test three self-authored interface shells—a shipment table, order cards, and a transfer form—with target-distractor text-box edge distances of 8, 40, or 96 pixels and DOM graph distances of zero or four element edges. Every combination receives ten trials per task and deployment ($N = 1,440$ across four deployments). The DOM variants are visually matched negative controls: the vision agent cannot observe their hidden structural distinction. A 240-trial natural-boundary control places Source and Reference in separate table columns, neighboring cards, or labeled form fields; its pixel and DOM distances are measured rather than independently manipulated. A further paired study ($N = 160$) compares inline-canvas baseline prompts with an explicit target-format instruction. It specifies five uppercase letters, a hyphen, and four digits, and excludes adjacent metadata without supplying the answer. The baseline and intervention share the same 730-pixel-wide canvas and 768×768 viewport.

Pixel distance is the minimum Euclidean edge-to-edge distance between browser text-range bounding boxes. DOM separation is the shortest parent-edge path between the elements containing the target and distractor text (zero when they share an element). Delimitation error is a nonempty candidate that contains the target with extra text or is a proper substring of the target. Candidates with other substitutions and empty candidates are recorded separately. For copying, the candidate is the last attempted typed string, while task success still requires an exact submitted value.

Calibrated, position-matched iframe rerun. A new 560-trial vision grid crosses four deployments, two tasks, seven unique depth/origin conditions, and ten repeats. Source and destination positions are randomized by task/repeat and paired across deployments, depths, and origins. Screenshots are 768×768 at device scale one; the prompt states those dimensions. Before model calls, pixel-center click/submit checks verify the action coordinates, and screenshots must be byte-identical across depths and origins at each paired position. These checks remove deterministic fixture scale and position differences without assuming perfect model localization. Geometry metadata remains scorer-only. All new arms retain the eight-action, 90-second budget; infrastructure failures are excluded and retried.

F.1 HISTORICAL COORDINATE AUDIT AND ORIGINAL-LAYOUT EXTENSION

Vision exposed the complementary failure mode (Figure 2B). It reached the canvas target in all 40 trials but completed none (0%; 95% CI 0–8.8%), and it completed all 120 shadow-containment trials. The original iframe completion rates varied non-monotonically despite 100% reachability, but inspection of all 40 anomalous `gpt-5.6-terra` screenshots and action logs shows that this pattern is not evidence of a depth mechanism. The application programming interface’s (API’s) high-detail preprocessing scaled the 1280×900 capture to approximately 1092×768 , while returned coordinates were executed in the original viewport. At depth 1, all 20 first copy clicks landed at $y = 224\text{--}226$, in the 12-pixel gap between the destination label and field. At depth 2, the same biased coordinate happened to land inside the associated label and focused the input, producing an accidental recovery. The depth-3 read target was fully visible, but each of the two condition-specific screenshots elicited the identical mis-transcription `AI PHA-7391` in all ten repeats. Repeats therefore sampled model behavior on identical pixels and fixed absolute positions. Appendix B retains the observed cells for transparency, but we treat them as a coordinate-runner audit and make no depth claim from them.

The two-deployment vision extension completed all 440 allocated trials with 440/440 reachable targets. Combined with the matched primary vision cells in Figure 2B, it shows that the original canvas failure is not universal across models: `gpt-5.4` completed 15/20 original-layout canvas

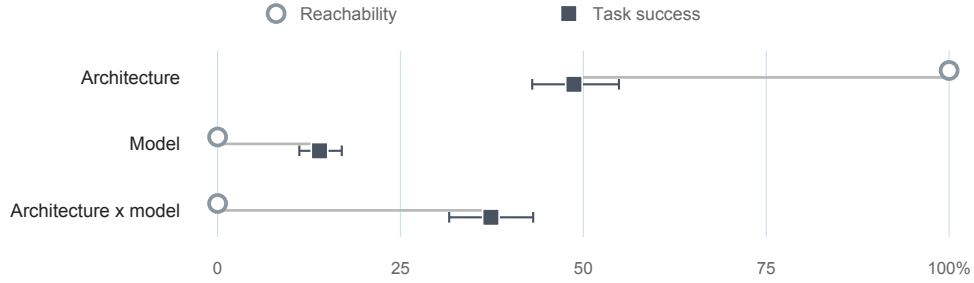


Figure 6: **Registered decomposition of the original grid, retained as a runner audit.** Points show each component’s share of the balanced two-way sum of squares; horizontal whiskers are stratified-bootstrap 95% intervals. Post hoc inspection identified coordinate-scale and absolute-position confounds in the vision cells that supply nearly all non-deterministic variation, so these shares are not interpreted as causal architecture, model-family, or depth effects.

trials, whereas gpt-5-mini completed 0/20, matching the two original deployments’ 0/20 each. The pooled four-model canvas rate is therefore 15/80, but the deployment-resolved result is primary because pooling masks this heterogeneity. Across all 11 conditions, gpt-5.4 completed 169/220 trials and gpt-5-mini 189/220. Their model-task-condition cells (Appendix B) remain position-sensitive: for example, gpt-5.4 copy success was 2/10, 0/10, and 7/10 at same-origin depths 1–3 while its read success was 10/10 throughout. The broader roster demonstrates model heterogeneity but does not rehabilitate a depth interpretation of fixed-position fixtures.

G REGISTERED VARIANCE DECOMPOSITION: DESCRIPTIVE AUDIT

The registered decomposition assigned 100% of reachability curve-shape variance to architecture. Task-success shares were architecture 48.7% (95% CI 43.0–54.9%), model 13.9% (11.2–17.0%), and interaction 37.4% (31.7–43.1%). These reproducible summaries describe the original stacks, not causal architecture, model-family, or depth effects. Structure-based agents were near ceiling outside canvas, leaving nearly all non-deterministic variation in vision cells affected by coordinate-scale and absolute-position confounds. Resampling identical screenshots cannot remove that bias. The original-layout extension’s 15/20 versus 0/20 canvas contrast broadens deployment coverage, but a vision-only extension cannot support a balanced architecture-model decomposition. That comparison requires calibrated, position-matched fixtures and the complete model roster crossed with all architectures.

H CALIBRATED IFRAME CELLS

The table reports the new calibrated and position-matched study. It does not replace the archived original cells in Table 4. S and X denote same- and cross-origin fixtures. The shared depth-zero cell is repeated for readability.

Table 5: Calibrated vision task success: successes out of ten trials per cell.

Deployment	Task	S0	S1	S2	S3	X0	X1	X2	X3
gpt-5.6-terra	Copy	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
gpt-5.6-terra	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
Kimi-K2.6	Copy	10/10	9/10	7/10	8/10	10/10	5/10	9/10	8/10
Kimi-K2.6	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
gpt-5.4	Copy	8/10	5/10	4/10	6/10	8/10	6/10	7/10	8/10
gpt-5.4	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
gpt-5-mini	Copy	1/10	2/10	3/10	0/10	1/10	2/10	1/10	1/10
gpt-5-mini	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10

All calibrated iframe geometry, initial screenshots, and click/submit checks are archived. Depth effects are reported as paired success-rate differences from depth zero, with bootstrap intervals in the artifact. The comparisons are exploratory; no equivalence margin was specified. Generalization availability is established from the independent text-range geometry checks; the original runner’s legacy reachability selector does not identify the new field containers. The calibrated leaf fixtures use a 360-by-44-pixel destination field and a new matched presentation. The original large-target field was at least 56 pixels high. This study estimates depth contrasts within the new fixtures; it does not isolate the effect of coordinate correction on historical success rates.

I NATURAL-BOUNDARY FIXTURE EXAMPLES

Visible boundaries in the natural-layout controls

A Separate table columns

Shipment details

Source	Reference
BRAVO-4826	CONTROL_COPY_VALUE_R5

Status: Ready | Service: Standard

B Neighboring cards

Order overview

Source	Reference
BRAVO-4826	CONTROL_COPY_VALUE_R5

Status: Ready | Service: Standard

C Labeled form fields

Transfer request

Source	Reference
BRAVO-4826	CONTROL_COPY_VALUE_R5

Service: Standard

Figure 7: Crops of the archived copy-task fixtures for the natural-boundary control. Source and Reference occupy separate columns, cards, or labeled fields. Tokens remain fixed; pixel distances and shortest DOM parent-edge paths are measured for every fixture. The complete viewport also includes the Destination form.

J ARTIFACT ACCESS AND REPRODUCTION REQUIREMENTS.

The code, benchmark fixtures, archived outcomes for the original and revision experiments, and analysis artifacts are available at <https://anonymous.4open.science/r/web-agents-go-blind-D87C/>.

The repository includes a lossless compact archive, a reconstruction script, and step-by-step reproduction instructions. Using Python 3.12 and the supplied pinned analysis dependencies, readers can reconstruct the archived files and regenerate the statistical results and result figures without browser installation or provider calls. Reconstruction verifies every restored file against its original SHA-256 checksum. Rerunning the live experiments additionally requires Chromium, the runner dependencies pinned in the lockfile, and access to the corresponding hosted model deployments. API credentials are supplied through environment variables and are excluded from the artifact